

TECHNICAL PROGRESS REPORT

Source-Linked Compression-Set Data for Silicone Formulations

Bryan V. Piguave

Brett M. Savoie

Department of Chemical and Biomolecular Engineering, University of Notre Dame
Notre Dame, IN 46556, USA

September 17, 2026

Executive summary

This report summarizes the work completed to assemble a source-linked record of silicone formulations and compression-set measurements from patents and journal articles. The practical question is whether a documented formulation and its test conditions can support useful guidance for seal design without requiring a new compression-set experiment for every candidate.

The central result is the funnel shown throughout this report: the search began with 2,925 downloaded documents. 1,930 contained at least one extracted property, 183 reported a compression-set value, and only 85 provided a value together with a composition and the three essential test conditions: temperature, time, and deflection. After source checking and curation, the current release contains 1,054 measurements over 605 formulations from 67 documents. This narrowing shows how much of the published record is sufficiently described to support quantitative comparison.

The release also distinguishes the material classes that are important for interpreting the data. A formulation is counted as a composite when it contains a dispersed filler role, and as a foam when it contains a blowing agent role. There are 525 composite formulations and 115 foam formulations. 110 formulations belong to both groups; the remaining formulations are 415 composite only, 5 foam only, and 75 neither.

The work produced a reliable baseline and a clear limitation. Test conditions alone give a leave-one-family-out R^2 of 0.289 with a mean absolute error of 17.5 percentage points. When entire source families are held out, the chemistry representations examined so far do not produce a statistically separable improvement. The current record is still too small, unevenly described, and concentrated in too few source families to establish a transferable chemistry effect, although chemistry may still matter in a larger and more balanced collection.

Taken together, these results define the present stage of the project. The work has moved beyond asking whether the literature contains silicone data at all. It now shows how much of that data can be made comparable, what types of formulations are represented, which parts of the test protocol are usually missing, and what level of prediction can be supported without overstating the evidence. The next phase can therefore be organized around specific gaps rather than around a general search for more papers.

1 What are the major goals of the project?

Compression set is a useful engineering property because it measures how much of an imposed deformation remains after an elastomer has been held under load for a specified time and temperature [1]. It is directly related to the long-term performance of seals, but it is also strongly dependent on the test protocol. Comparing a value measured after 22 h at 175 °C with one measured after 70 h at 150 °C requires accounting for those different temperatures and durations.

The first goal is therefore to gather measurements together with the conditions that make them interpretable. The second is to preserve the chemistry of the formulation: base polymer, crosslinker, catalyst or initiator, fillers, additives, and any blowing agent. Silicone elastomers are formulated network materials, and their behavior depends on the network created by the cure chemistry, the interaction with fillers, and the processing and aging history [2–4].

The third goal is to determine what can reasonably be learned from the published record. Polymer formulations are difficult to model because the design space is large while much of the practical knowledge is recorded in qualitative descriptions, patents, and tables with inconsistent terminology [5–7]. The project therefore asks two separate questions: what information can be recovered and checked, and how much predictive value does the recovered chemistry add beyond the test conditions themselves?

These goals are deliberately connected because a prediction becomes useful for engineering only when its numerical value can be traced to a source. Without a clear distinction between a filled elastomer and an expanded foam, a chemistry label can make a model appear more general than it is. Conversely, a carefully curated but very small record may be unable to demonstrate an effect that is chemically real. The project treats source quality, material classification, and prediction as parts of the same research problem.

The project is aimed at a dependable foundation for deciding which comparisons are meaningful and which experiments or source acquisitions would most improve the evidence. In that sense, a clear negative result about the present corpus is useful because it prevents a high apparent score from being mistaken for reliable transfer to a new material family.

2 What was accomplished under these goals?

The project established a complete path from a broad literature record to a curated set of comparable measurements. Documents were collected from patents and journal literature, screened for silicone materials and quantitative compression-set results, and read in the context of the surrounding formulation and testing descriptions. Duplicate source families were treated as one scientific source so that repeated publication of the same work did not create artificial evidence.

The completed work has two complementary outputs. The first is the research record itself: a set of formulations and measurements that can be inspected against the pages from which they came. The second is an account of the limits of that record. It identifies the points at which a document stops being quantitative, the material classes that are present in the surviving formulations, and the kinds of uncertainty that remain after careful reading. Together, these two outputs support collaboration: the record supports analysis, while the limitations identify where expert judgment or new experiments are still needed.

Figure 1 summarizes how the record is reduced to a defensible comparison. Its top panel makes the funnel explicit: each additional requirement removes documents that do not provide enough information for a defensible comparison. The lower-right panel records the material classes in the final release. These classes overlap by design, because a filled silicone foam is both a composite and a foam.

2.1 Major Activities

The work had four connected activities. First, the source record was expanded without requiring the property to appear in a title or abstract. This was important because many patents present compression-set values only in example tables. Second, formulations were read as recipes rather than as isolated chemical names.

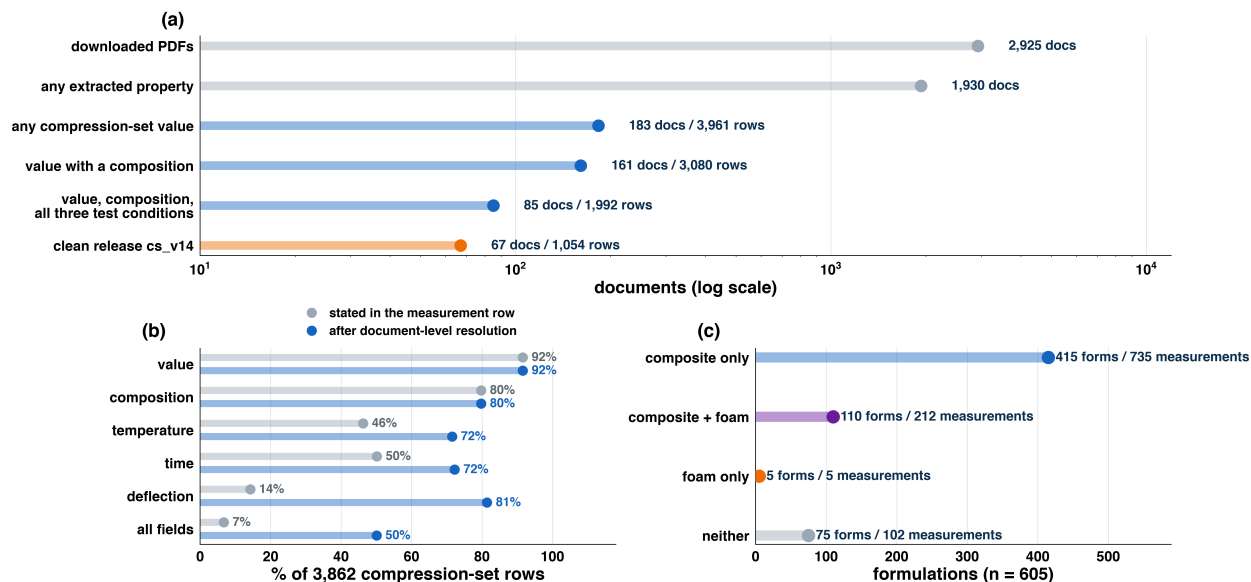


Figure 1: The record narrows from documents to usable measurements. (a) Documents remaining as composition, test protocol, and release-quality requirements are applied; row counts are shown where the stage is defined on measurements. (b) Information stated in the measurement row itself compared with information recovered from the document as a whole. (c) Formulations classified as composite, foam, both, or neither. Composite means a dispersed-filler role; foam means a blowing-agent role.

Trade names, component labels, filler descriptions, cure systems, and amount conventions were retained when they were part of the source’s description.

2.1.1 Source discovery and screening

The source search was designed to favor recall at the beginning and judgment later. Silicone terminology varies widely across patents and articles, and a document may report compression set only in a long example section. The search therefore gathered documents using broad material and application language before asking whether a document actually contained the target property. This approach retained documents that would have been lost if the property name had been required in the title or abstract.

The screening decision also had to distinguish silicone as a matrix from silicone as an additive, modifier, crosslinker, or minor component in another polymer. This distinction matters scientifically because a polyurethane containing a silicone additive may be a valuable material, but it does not answer the same question as a silicone elastomer whose network carries the load. The current record focuses on the latter so that the measured property has a consistent material context.

2.1.2 Formulations as complete recipes

The second activity was to read each formulation as a complete recipe. A source may identify the base polymer in one paragraph, describe a filler in a table, and identify the cure system in a separate preparation step. Those pieces cannot be treated as independent observations. They describe one material and must remain together if the result is to be interpreted.

The same principle applies to anonymous labels. The label “Component A” acquires chemical meaning only when the document defines it; in one example, the source identifies it as a vinyl-terminated polysiloxane with a stated viscosity and identifies Component B as a hydride-containing crosslinker. What matters is the definition supplied by the document, while the label alone carries little information. Trade names and generic descriptions were retained when they carried information about the formulation, and uncertain identities were

kept visibly uncertain rather than replaced with a convenient guess.

Third, test conditions were connected to the measurements to which they belong. Temperature, time, deflection, medium, specimen geometry, and stated test standard were kept separate because each can change how a compression-set value should be interpreted. When a source stated conditions in a methods paragraph rather than beside the result, the conditions were connected back to the relevant measurements only when the source supported that connection.

This step is especially important for aging studies. A source may describe a pre-aging exposure, a cure schedule, and a compression-set hold in the same experimental section. These events are related, but each requires a separate field in the record. The record keeps them distinct so that a future user can ask whether two measurements used the same aging history rather than assuming that every temperature mentioned in a document was the test temperature.

Fourth, the resulting records were checked against the source pages. A stratified audit found the released compression-set value correct in 96.8% of checkable cases, with a Wilson 95% interval of 94.7–98.1%. Temperature and deflection were correct in every checked case. The audit also identified specific table-reading failures and source-family duplicates that would have been difficult to detect from numerical consistency alone.

Verification was treated as a scientific result in its own right. When a value disagreed with a source page, review considered both its numerical plausibility and the alignment of the formulation row, specimen, and stated test conditions. This distinction matters because a shifted table can produce a chemically reasonable formulation with the wrong amount, or a reasonable value with the wrong protocol. The audit therefore examined the relationships among fields as well as the fields themselves.

2.2 Specific Objectives

Objective 1 – Build a quantitative source record. The project converted a dispersed set of patents and articles into a collection in which each measurement remains connected to its source document, formulation, and test description.

This objective is important for more than citation. Source linkage allows a future reader to check whether a number was reported for a neat elastomer, a filled material, a foam, or a multi-part system. It also allows a correction to remain local. If a source later clarifies a cure condition or a component identity, the affected result can be revisited without silently changing unrelated measurements.

Objective 2 – Distinguish material classes. The project now reports which released formulations contain dispersed fillers and which contain blowing agents. This makes the composite and foam populations visible instead of combining them into one undifferentiated silicone category.

This objective responds directly to the intended use of the record. A filler can change modulus, heat transfer, interfacial adhesion, and recovery, while a blowing agent changes the internal structure and the way a specimen carries load. Both effects can appear in a compression-set result, but they should not be interpreted as the same kind of chemistry. Reporting the overlap makes the classification useful without pretending that the categories are exclusive.

Objective 3 – Establish a defensible predictive baseline. The project compared condition-based prediction with several broad descriptions of chemistry while keeping source families separate during evaluation. The purpose was to identify what the current evidence supports, not to maximize a single score by mixing related examples between training and testing.

The baseline provides a reference point for future work. If a new source collection or a new chemical description is added later, its value can be judged against a stable condition-based comparison. This is more informative than reporting a single best score because it shows whether a proposed improvement comes from chemistry or from the model recognizing the source document in which the measurement was originally reported.

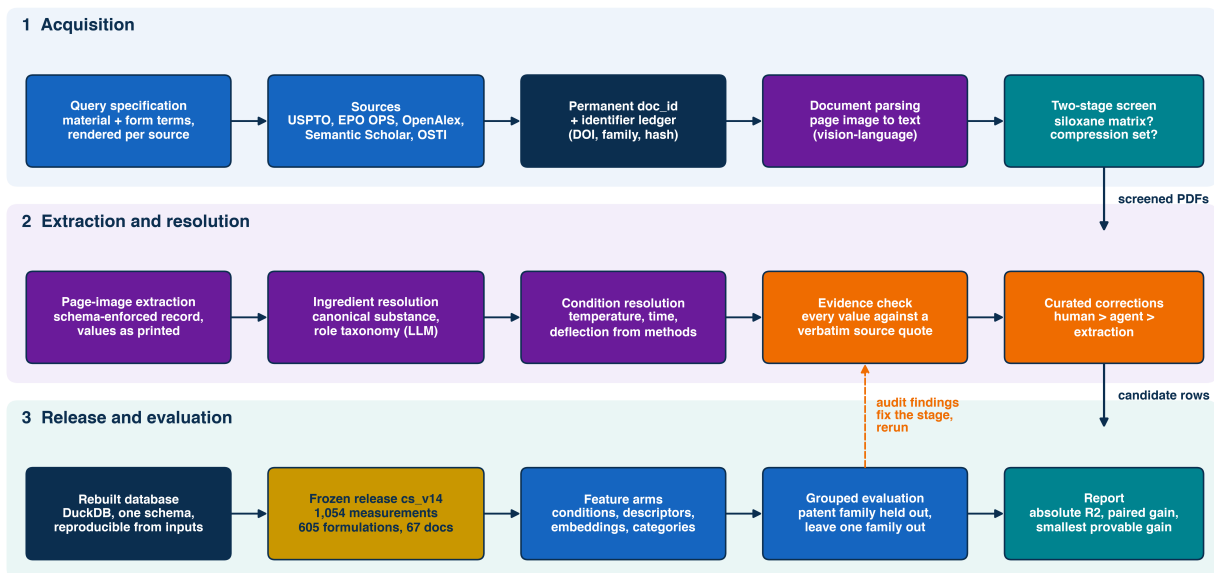


Figure 2: High-level progression of the work. The project moves from source discovery, through formulation and test-condition reading, to source checking, material classification, and evaluation. Each stage preserves the connection between a reported value and the document in which it appeared.

3 Significant Results

3.1 The document funnel identifies the quantitative core

The funnel begins with 2,925 downloaded documents and ends with 67 source documents in the current release. In between, the record passes through 1,930 documents with any extracted property, 183 documents with a compression-set value, 161 documents with a stated composition, and 85 documents with a complete test protocol. The corresponding compression-set row counts are 3,961 for the broad property population, 3,080 with a composition, and 1,992 with all three conditions.

This sequence is the most important result for planning future work. A large number of documents does not automatically create a large modeling dataset. The limiting resource is the number of source families that report a complete recipe, a measured value, and enough test information for comparison. The funnel shows exactly where additional reading and curation can increase the usable population.

The first reduction is broad: many documents mention a material or a property without presenting a numerical measurement that can be used. The second reduction is more consequential for formulation science. A compression-set number without a recipe cannot be connected to a base polymer, filler, or cure system. The third reduction concerns the protocol: even a complete recipe and a measured value remain difficult to compare when the temperature, time, or deflection is absent or ambiguous.

The final reduction is a judgment about whether the surviving record is complete enough to support a scientific claim. It removes cases that are prophetic rather than measured, outside the physical range of the property, duplicated across source families, or too incomplete to identify the material being tested. This last step is intentionally conservative because a smaller set of traceable results is more useful for collaboration than a larger set whose composition or test conditions cannot be defended.

3.2 The release separates composites and foams

The material classification is based on the ingredient roles supported by the source. In this report, a composite contains at least one reinforcing, functional, extending, or lightweight filler, whereas the foam

label denotes a formulation with a documented blowing-agent role. The four categories below partition the 605 released formulations, while the aggregate composite and foam populations overlap.

Table 1: Operational material classes in release cs_v14. The composite and foam labels can overlap; the four rows partition the formulations.

Formulation class	Formulations	Compression-set measurements
Composite only	415	735
Composite and foam	110	212
Foam only	5	5
Neither composite nor foam	75	102
Total	605	1,054

In aggregate, 525 formulations from 60 documents are composites and 115 formulations from 17 documents are foams. The overlap is substantive rather than an accounting error: the 110 formulations in both classes contribute to both totals. This distinction should be carried into future comparisons because foam structure, filler content, and cure conditions can affect compression set in different ways.

The classification is deliberately based on what the formulation contains, not on an attempt to infer every product name used by the source. A document may call a material a sponge, cellular rubber, or lightweight elastomer while the formulation record uses a different vocabulary. The blowing-agent role is the most consistent evidence available across sources, and the filler roles provide a similarly consistent basis for the composite label. Sources that do not support either label remain in the release as neither rather than being forced into one of the two populations.

At the document level, composites occur in 60 source documents and foams in 17 documents. The overlap in formulations is useful for future analysis because it creates a natural question: does the compression set of a filled foam follow the behavior of filled solid elastomers, the behavior of foams, or a separate regime? The current report identifies the population needed to ask that question without claiming that it has already been answered.

3.3 The current release is broad but uneven

The current release cs_v14 contains 1,054 measurements over 605 formulations from 67 documents, including 65 patents and 2 journal articles published between 1974 and 2026. The median compression set is 25% and the interquartile range is 13–51%. The sources use several amount conventions and describe cure systems, fillers, and test geometries with different levels of detail.

The release is consequently best understood as a map of the evidence rather than as a uniform catalog. Some source families provide many formulations and many aging conditions; others provide one or two comparative values. Some sources name a commercial material in detail, while others provide only a generic description. These differences affect which chemistry comparisons are possible and how much confidence can be placed in a trend; they are therefore more than cosmetic variations in presentation.

Figure 3 summarizes this variation across source documents and testing conditions. The measurements are unevenly distributed across source documents, and the test conditions cluster around common industrial protocols while extending to other temperatures and times. This unevenness is why the report emphasizes source families and document-level comparison rather than treating every row as an independent piece of evidence.

The field-coverage information in the figure also provides a guide for future reading. Temperature and time are often recoverable from a methods section, whereas specimen geometry and the exact standard may remain absent. A future release can therefore be improved in two ways: by adding entirely new source families and by completing the protocol information for documents that already contain useful measurements. The funnel does not distinguish these two kinds of progress by itself, so both should be reported separately in later updates.

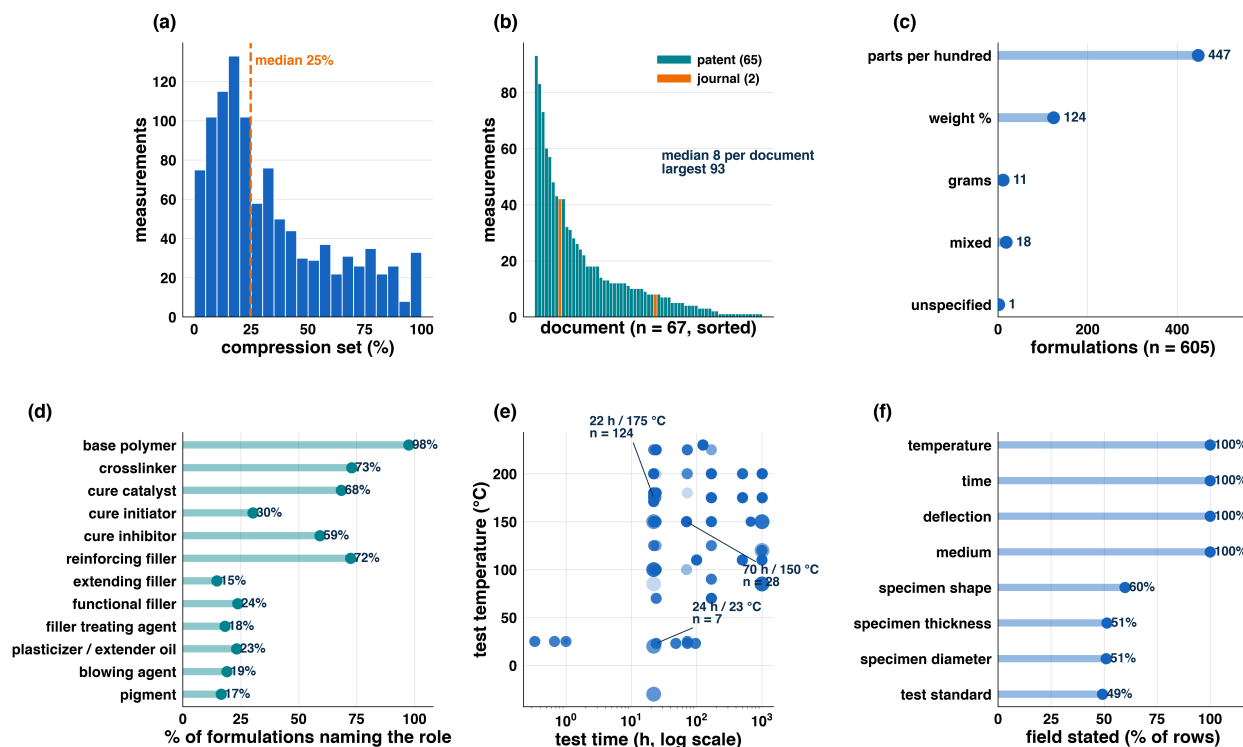


Figure 3: What the current release contains. The release includes measurements spanning a wide range of compression-set values, source documents, formulation amount conventions, ingredient roles, and test protocols. Empty fields are retained as evidence that the source was silent; they are not silently filled by an assumed value.

3.4 What the release reveals about experimental practice

The source record shows that compression-set testing is not reported in one uniform way. Some studies are organized as systematic formulation series in which one filler or cure variable changes at a time. Other studies compare commercial materials, product families, or processing histories with less detail about the underlying chemistry. Both types of study can be valuable, but they support different conclusions. A controlled series is useful for studying a local formulation effect, whereas a product comparison may be more representative of an application but less informative about mechanism.

The same distinction appears in the amount conventions. Recipes may be given relative to the base polymer, as percentages of the final mixture, in grams, or with only a qualitative statement that a component was included. These conventions are not interchangeable without assumptions. Retaining the original basis preserves the source’s meaning and makes it possible to select only the comparable subset for a particular analysis.

The test protocol is similarly heterogeneous across documents. Common industrial holds create useful concentrations of data, but they can also make the literature appear more consistent than it is. Room-temperature measurements, long-duration aging, post-cured specimens, oil exposure, and different specimen geometries may represent different use cases rather than noisy versions of one test. The release keeps these distinctions visible so that future comparisons can be organized around an application or a protocol instead of averaging unlike measurements together.

This observation changes how the dataset should be used. The release is well suited to identifying repeated protocols, locating formulation families, and selecting candidate comparisons. It should serve as a starting point for an agreed qualification protocol rather than as a substitute for one. A future experimental campaign can use the release to choose formulations and then measure them under a common protocol, thereby separating formulation variation from differences in how the test was run.

3.5 A conditions-only baseline is presently the defensible result

The first predictive comparison used the test conditions as the common basis for every formulation. Additional chemistry descriptions were then tested to ask whether they improved transfer to source families that had not been used to fit the prediction. Conditions alone reached a leave-one-family-out R^2 of 0.289 with a mean absolute error of 17.5 percentage points.

This baseline has a practical interpretation because the test protocol is one of the strongest determinants of the reported number, rather than a minor annotation attached after the chemistry. A model that knows the aging temperature, time, and deflection has information about the severity of the test even when the source gives only a limited formulation description. That makes the conditions-only result a useful floor for judging whether additional chemistry information is helping.

Adding stated viscosity, ingredient descriptions, combined descriptors, or ingredient categories changed the leave-one-family-out score by small amounts whose uncertainty intervals included zero. The best chemistry representation reached 0.280, but the current corpus cannot establish that this is better than the conditions-only baseline. Random-row evaluation gives a substantially higher diagnostic score, including 0.737 for ingredient categories, because related examples from the same document remain on both sides of the comparison.

The difference between the two evaluation settings is itself informative. A random-row comparison can use the fingerprint of a document: formulations from one patent or article often share a base polymer, cure system, terminology, and testing habits. Holding out the source family asks the harder question of whether the same description can guide a prediction for a new source. The present record contains enough within-document regularity to support the first task, but not enough independent families to demonstrate the second.

Figure 4 presents this result as a model ladder. The main conclusion is a statement about evidence: the present release supports a useful condition-based baseline, while evidence for a chemistry improvement across unseen source families remains insufficient. A larger and more balanced set of complete source families, or a trusted validation set, is needed to answer that question.

The result sets a useful standard for the next stage. A new chemistry representation should improve the unseen-family comparison, remain stable when individual source families are removed, and make a difference large enough to affect an engineering choice. Until those conditions are met, the responsible conclusion is that the current release has not yet separated formulation chemistry from source and protocol variation. This limitation concerns the evidence currently available, rather than the possibility that chemistry affects the property.

3.6 How the result should guide experiments

The present result suggests a staged experimental strategy. In the short term, the most valuable measurements will connect a trusted formulation to a clearly specified protocol and fill a gap in the current material classes. For example, a small matched set of composite and foam formulations tested under the same conditions could be more informative than a large collection in which each source uses a different test history.

The next stage should also separate screening from qualification. The literature record can help identify candidates and reveal which ingredients or material classes recur across sources. Qualification measurements should then be run under a common protocol selected for the intended application. Keeping those purposes separate prevents a screening result from being treated as a direct replacement for an engineering test.

This staged approach gives the collaboration a practical decision rule. If a new source adds a material class or protocol that is currently missing, it may be valuable even before its data improve a global score. If a new experiment repeats a well represented source family without clarifying the engineering threshold, it may add less information. The funnel, class labels, and condition-based baseline together provide the context needed to make that choice.

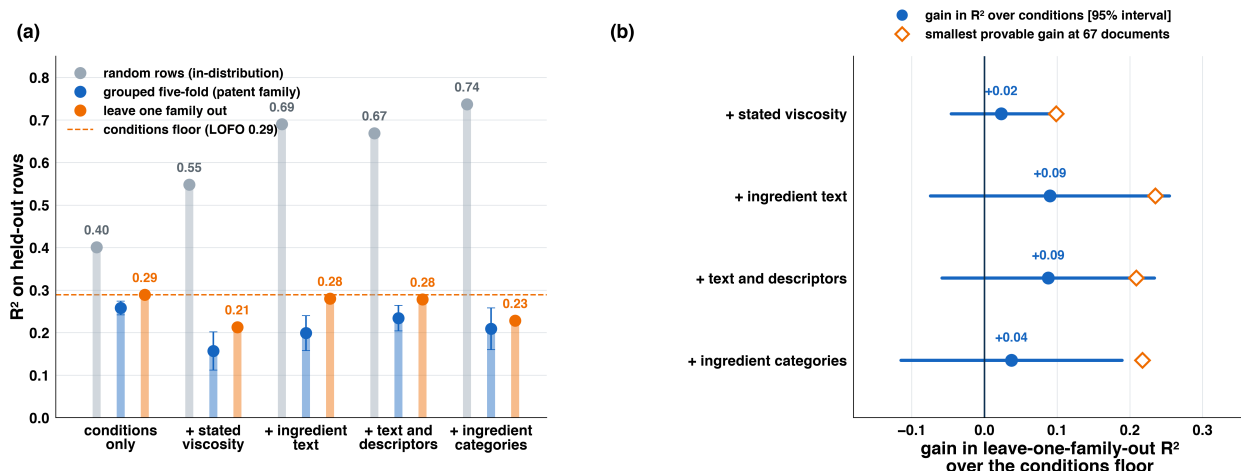


Figure 4: The predictive baseline and the chemistry question. The conditions-only baseline is compared with several broad chemistry descriptions under familiar-document and unseen-family evaluations. The chemistry gains remain within the uncertainty supported by the current number of source families.

4 Lessons from verification

Several verification cases changed the way the record is interpreted. A correct numerical value can still be attached to the wrong formulation when two members of a patent family present different tables. A plausible chemical name can still be wrong when a source uses a similar-looking end group. A table can appear complete while a shifted column has assigned an amount or a test condition to the neighboring row. These cases show why numerical agreement, chemical plausibility, and direct source checking are all needed.

The most consequential lesson is that source context matters more than the title of a document. Some documents that appear general from their title contain the most useful aging data, while some documents that mention silicone do not describe a silicone matrix and should not be included. The funnel and the audit are therefore complementary: the funnel shows how many documents remain, while the audit tests whether the surviving values still mean what the source reported.

Four recurring failure modes are especially important. First, a reprinted patent family can preserve the same property values while presenting a different or incomplete formulation table. Second, a chemical name can be plausible but still disagree with the source in a way that changes reactivity. Third, a table can shift a number into a neighboring column without producing an obviously impossible value. Fourth, a document can contain a complete recipe but report a result under conditions that are described only elsewhere or that apply to a different specimen. None of these errors is reliably detected by looking only at the final numerical range.

The verification process therefore changed the project’s definition of a usable result. A usable result combines a reported value with enough formulation, specimen, and protocol context that another reader can understand what was measured. In some cases the appropriate outcome is to keep the value but mark a missing field. In other cases the right outcome is to leave the value out until the source can be resolved. This distinction keeps uncertainty visible instead of turning it into false precision.

These lessons should also guide future acquisition, because new documents should be judged by the quality and diversity of the formulations they add, not only by the number of compression-set rows. A source with one carefully described composite or foam may be more valuable than a source with dozens of repeated examples whose chemistry and protocol cannot be separated.

5 Companion benchmark and broader value

The table-reading work was also tested on a 1968 reference volume containing tabulated organosilicon properties. That setting offered a fixed ground truth against which names, structures, values, and pressures could be checked. The verified benchmark contains 509 measurements over 436 structures from 159 references. The benchmark was useful because it separated errors in reading a table from uncertainty in interpreting a modern patent formulation.

The broader contribution is a change in the unit of analysis. Silicone formulation development is often discussed in terms of individual ingredients or isolated property values. The present record connects the source document, the complete formulation, the material class, and the test protocol. That connection makes it possible to ask which evidence is genuinely comparable and which apparent trend is only a consequence of how a document was written.

The companion benchmark also clarified the role of evaluation. In a reference volume, the true answer is available for every selected table entry, so a reader can distinguish a transcription error from a disagreement caused by chemical interpretation. In the formulation record, by contrast, some disagreements reflect the source’s own ambiguity. The benchmark supplied a controlled setting for improving reading and verification practices before those practices were used to make claims about modern formulations.

The value of this connection extends beyond compression set. The same record can support questions about aging, hardness, tensile behavior, cure response, or other properties when the source provides them. The material-class labels also create a starting point for comparing dense composites with cellular materials. The current report focuses on compression set because it is the property for which the protocol and source coverage are presently most mature.

6 Changes in approach and reasons for change

The work began with the expectation that a large literature search would quickly produce a large modeling table. The evidence showed that this assumption was too simple. Many documents contain a property value but not a complete formulation, and many complete recipes do not provide the test conditions needed to compare the value with another source. The project therefore shifted emphasis from counting retrieved documents to measuring the progressive loss of information through the funnel.

The second change was to treat the source document as the unit of context. Early comparisons can make several rows from one patent appear to be many independent experiments. In reality, those rows often share the same base polymer, preparation history, terminology, and testing convention. Holding out source families makes the prediction question more demanding, but it also matches the intended use: predicting a material described in a source that was not already part of the evidence used to establish the relationship.

The third change was to make material classes explicit. The initial release could have reported one population of silicone formulations, but that would have hidden an important distinction for the collaborators. A filled solid elastomer and a cellular foam can both be described as silicone rubber while having different structures and different responses to load. The composite and foam labels are therefore included in the report as a first classification layer, with the overlap shown rather than suppressed.

These changes made the report more conservative but also more useful. The project now explains what result it can produce and why a larger or more detailed source record is needed before stronger chemistry claims can be made.

7 Actual or anticipated problems and actions to resolve them

The central anticipated problem is uneven source coverage. The current release contains many examples from a limited number of source families, while some silicone cure systems and product forms remain sparsely represented. Additional rows from an already well represented document may improve interpolation while doing

little for transfer to a new family. Future acquisition will therefore track both the number of measurements and the diversity of source families.

A second problem is ambiguity in formulation terminology. Sources use names such as silicone rubber, silicone gel, sponge, cellular rubber, and elastomer in ways that are not always consistent. The report addresses this by relying on the formulation evidence for the composite and foam labels, but some cases will still require expert review. The next stage will record the source’s own product description alongside the operational classification so that the two are not confused.

A third problem is incomplete test context. Temperature and time can often be recovered from nearby methods, but deflection, specimen geometry, medium, and post-cure history may remain absent. Those missing fields limit comparisons even when the value and recipe are clear. The action is to preserve the missingness, prioritize documents where a complete protocol is likely to be recoverable, and request trusted reference conditions from collaborators.

A fourth problem is that a prediction error may not have the same engineering meaning for every application. A difference that is negligible for screening may be important for a long-life seal. The project therefore needs an application-level error threshold before it can decide how much additional data is enough. A collaborator’s statement of the error that would change a design decision will make the next collection effort more targeted.

The project has no reason to treat these problems as reasons to stop. They are the information needed to decide what to do next. The funnel identifies the missing source evidence, the class labels identify the missing material comparisons, and the baseline identifies the predictive improvement that has not yet been demonstrated.

8 What is the impact on society beyond science and technology?

Silicone seals and gaskets are used in applications where failure can create maintenance, safety, and reliability costs. Better organization of the evidence behind compression-set measurements can help engineers compare materials more carefully and identify when a published value does not describe the same test as the proposed application. The immediate project does not claim to replace qualification testing. Its contribution is to make the literature a more reliable starting point for deciding which materials and tests deserve that investment.

The project also illustrates a broader practice for scientific data. A large collection is not automatically a useful collection. Showing what is removed by each requirement makes the limitations of a data resource visible to the people who may rely on it. That transparency can reduce overconfident use of models and make later improvements easier to evaluate.

9 What opportunities for training and professional development has the project provided?

This project has provided training at the intersection of polymer chemistry, literature analysis, data curation, and scientific communication. The work requires moving between a source page and the chemical meaning of a formulation, then explaining why a particular value can or cannot be compared with another value. That combination is different from either conventional literature review or conventional modeling.

The work has also emphasized judgment and uncertainty. Researchers must learn to recognize when a source is specific enough to support a claim, when a generic description is still useful, and when an apparently complete table is unreliable. They must communicate the curation decision together with the evidence that motivated it. This is an important skill for materials data work because the most serious errors are often plausible rather than obviously numerical.

The composite and foam classification provides a concrete training example. It requires recognizing the function of an ingredient, distinguishing a blowing agent from a filler, and explaining why the two populations can overlap. It also gives collaborators a common vocabulary for reviewing future formulations

and identifying which source descriptions need expert attention.

10 What is the impact on the principal discipline?

The project contributes to polymer and elastomer informatics by treating a formulated material, rather than an isolated repeat unit or ingredient, as the unit of analysis. This is important for silicone systems because the measured property emerges from a network containing a base polymer, a cure system, fillers, additives, and a processing history. A representation that keeps only the polymer name cannot preserve all of the factors that distinguish two commercially meaningful formulations.

The project also makes the limits of the literature measurable. Instead of reporting only the number of documents found, it reports how many survive the requirements for a composition and a complete test protocol. That distinction can guide future data-collection efforts in elastomers and in other formulated materials where useful information is spread across tables, examples, and methods sections.

Finally, the work establishes a benchmark for claims about chemistry. A chemistry representation should be judged by how well it transfers to new source families, rather than by how well it interpolates among examples from a source it has already seen. This framing is directly relevant to the use of data-driven methods in polymer design.

11 What is the impact on other disciplines and potential users?

The record is relevant to chemical engineering, materials science, reliability engineering, and scientific data management. For chemical engineers, the record connects formulation variables to a property used in seal and gasket design. For materials scientists, it makes the distinction between dense filled elastomers and cellular materials visible. For reliability engineers, it emphasizes that the test protocol is part of the evidence rather than an afterthought.

The work may also be useful to groups that are building data resources for other formulation classes. The funnel provides a simple way to separate a large discovery population from a smaller quantitative core. The source-linked approach provides a way to preserve the context needed for later review. The classification approach demonstrates how a material category can be defined from evidence in a recipe while still allowing overlapping categories.

The immediate users of this report are collaborators who need to decide which silicone families and test regimes to study next. The report is intended to make that decision concrete. It shows what is already represented, where the record is thin, and why additional rows from the same source family may be less valuable than a smaller number of complete formulations from a new family.

12 How have the results been disseminated?

The results are being organized as a technical report, a frozen research release, source-linked records for the collaborating group, and a documented set of figures that communicate both the size of the record and its limits. The funnel is intended to make acquisition priorities visible to collaborators: new documents are valuable when they add complete formulations and complete test conditions, not merely when they increase the number of downloaded papers.

The report also provides a common language for discussing material classes. Future meetings can distinguish filled silicone composites, silicone foams, and formulations that belong to both categories instead of comparing all of them as one population.

The funnel and the classification table are designed for communication as much as for record keeping. They give collaborators a compact way to see why the current release is smaller than the starting literature set and what kind of formulation information is present in the result. This is particularly useful when

deciding whether a new source should be prioritized for its property values, its chemistry, its test protocol, or its representation of a material class that is currently underrepresented.

The report also preserves the distinction between what is measured and what is inferred. A source-linked measurement can be shared as evidence. A proposed material label or chemistry interpretation can be shared as a useful working description while remaining open to review. Keeping those two levels separate will make it easier to incorporate expert feedback without rewriting the history of the source record.

13 What do you plan to do next to accomplish the goals?

The next stage has three priorities for expanding and testing the release. First, confirmed errors from the audit will be corrected and additional source families will be added through the same source-linked process. Second, the composite and foam labels will be reviewed alongside cure system and base-polymer information so that the material classes can be compared without conflating structure and chemistry. Third, the predictive baseline will be reevaluated after the record is expanded and after collaborators provide a small validation set with trusted formulations and test conditions.

Two inputs from collaborators would have immediate value. A set of formulations that Los Alamos trusts, with or without measured values, would provide a fixed reference for evaluating extraction and representation choices. Also needed is a statement of what prediction error would change a design decision at a named test protocol. That engineering threshold would determine whether the next effort should focus on more source families, more complete chemistry descriptions, or targeted measurements of composites and foams.

The acquisition effort should also be organized around balance. The current record contains many examples of some cure systems and material classes and fewer examples of others. Future searches should deliberately seek source families that add missing combinations of base polymer, cure chemistry, filler content, foam structure, and test regime. The aim is to make comparisons less dependent on a few documents while expanding the total number of measurements.

For composites, the next questions concern the type and amount of dispersed phase, surface treatment, and the way filler content interacts with cure and aging. For foams, the next questions concern the blowing-agent system, the resulting cellular structure, density where reported, and whether the test specimen is representative of the intended application. These questions can be addressed only when the relevant source descriptions are retained together with the compression-set result.

The next report should therefore include both an expanded release and a clear account of what changed: newly added source families, newly resolved conditions, corrections to earlier records, and the number of formulations in each material class. Reporting the changes this way will make progress visible even when the predictive score does not change immediately.

14 Overall assessment

The project has reached a useful intermediate point. The literature search has produced a broad starting population, while the successive screening steps have identified the smaller group that can support a defensible comparison. The funnel is therefore both a result and a guide for the next phase. The central challenge is to find sources that report a formulation, a measured compression-set value, and enough test conditions for the value to be interpreted correctly.

The material-class analysis adds another layer of meaning to that result. Of the 605 formulations in the current release, 525 are classified as composites and 115 as foams. Because 110 formulations belong to both groups, the classes should not be treated as mutually exclusive alternatives. The release contains 415 composite-only formulations, 110 formulations that are both composites and foams, 5 foam-only formulations, and 75 formulations that are neither under the current operational definitions. This distribution makes the current emphasis clear: composite behavior is better represented than foam behavior, while the overlap between the groups must remain visible in future analyses.

The same caution applies to the testing record. A reported compression-set percentage is not sufficient on its own to define a comparable observation. Temperature, duration, deflection, specimen details, and post-cure history can change the meaning of the result. The most useful records are consequently those that preserve the formulation and the test description together. This principle explains why the final release is smaller than the set of documents identified at the beginning, but also why the smaller release is more useful for engineering interpretation.

Taken together, the present work establishes a transparent baseline for expansion. Future additions can be judged by the stage at which they enter the funnel, the material class they strengthen, the completeness of their test conditions, and whether they add a genuinely new source family. That reporting structure will keep the project focused on evidence that can support design decisions rather than on document counts alone.

The stages of the funnel also distinguish several different kinds of progress. Moving from the downloaded document set to documents with an extracted property measures whether the literature search has reached technically relevant material. Moving from those documents to records with a compression-set value measures whether the desired property is actually reported. Requiring a formulation then changes the question from “What value was observed?” to “What material produced the value?” Finally, requiring the complete set of conditions changes the question to “Can this observation be compared with another one?” These are different accomplishments, and combining them into a single document count would hide where the main limitations occur.

The funnel therefore provides a practical way to prioritize review. A source that reaches the composition stage but lacks one test condition may be worth revisiting if that condition is central to an engineering comparison. A source that reports a value but no usable formulation may be valuable as background evidence, but it should not be allowed to carry the same weight as a complete record. Similarly, a new document that adds a foam formulation may be more important to the scientific balance of the release than several additional composite measurements that repeat an already well-represented condition.

This perspective is especially important for the composite and foam categories. The word “composite” describes the presence of an added phase in the recipe, whereas “foam” describes a cellular structure or a formulation containing a blowing-agent role. A formulation can satisfy both descriptions, as the current overlap demonstrates. Keeping the two labels independent preserves the ability to ask whether a measured change is associated with filler content, cellular structure, or both. It also prevents the report from treating foams as a separate universe that has no relationship to the filled elastomers elsewhere in the release.

The categories that fall outside both definitions are informative as well. They include formulations that may be unfilled dense elastomers, formulations whose ingredients are not resolved sufficiently to assign a role, and records that do not belong to either targeted class. These formulations provide a useful comparison population. They can help identify whether a trend is specific to an added phase or cellular structure, or whether it is already present in the base material and cure system. Retaining them avoids narrowing the release so much that every observed difference is forced into a composite-versus-foam explanation.

For this reason, the current release should be interpreted as a structured starting point rather than a final map of silicone performance. Its strongest contribution is that it makes the boundaries of the evidence visible. The source count, row count, condition count, and class count answer different questions, and each should remain available when the record is expanded. This will allow future reports to show how the collection grew, whether it became more complete and balanced across material classes, and whether it became more useful for comparing formulations under the same test conditions.

15 Information retained in the research release

Each released measurement remains connected to its source document and to the formulation from which it was reported. The retained information includes the ingredient descriptions and roles, amounts and amount conventions, measured compression set, test temperature, time, deflection, and any stated medium, specimen geometry, or post-cure history. Missing information is left visible as missing so that future users can distinguish a source that did not report a quantity from a formulation that did not contain or require it.

The current release is a snapshot of the evidence available at this stage of the project. Its value lies in the traceability that allows a future correction, new material class, or new test condition to be connected back to the page on which the original claim was made, as well as in the number of rows it contains.

The release should be read with the same care as the sources themselves. A filled cell means that the source provided usable information; an empty field means that the source did not provide it or that the information could not be resolved with sufficient confidence. This convention protects later users from mistaking an assumption for an observation and makes the remaining work easy to describe.

References

- [1] ASTM International. ASTM D395: Standard test methods for rubber property—compression set. ASTM International, West Conshohocken, PA. Verify current revision year.
- [2] Piotr Stanislaw Mazurek, Sindhu Vudayagiri, and Anne Ladegaard Skov. How to tailor flexible silicone elastomers with mechanical integrity: A tutorial review. *Chemical Society Reviews*, 48:1448–1464, 2019. doi: 10.1039/C8CS00963E.
- [3] Yinglei Yue, Hui Zhang, Zhong Zhang, and Yunfa Chen. Tensile properties of fumed silica filled polydimethylsiloxane networks. *Composites Part A: Applied Science and Manufacturing*, 54:20–27, 2013. doi: 10.1016/j.compositesa.2013.06.016.
- [4] Mogon Patel, Mark Soames, Anthony R. Skinner, and Thomas S. Stephens. Stress relaxation and thermogravimetric studies on room temperature vulcanised polysiloxane rubbers. *Polymer Degradation and Stability*, 83(1):111–116, 2004. doi: 10.1016/S0141-3910(03)00231-3.
- [5] Debra J. Audus and Juan J. de Pablo. Polymer informatics: opportunities and challenges. *ACS Macro Letters*, 6(10):1078–1082, 2017.
- [6] Keith T. Butler, Daniel W. Davies, Hugh Cartwright, Olexandr Isayev, and Aron Walsh. Machine learning for molecular and materials science. *Nature*, 559:547–555, 2018.
- [7] Rampi Ramprasad, Rohit Batra, Ghanshyam Pilania, Arun Mannodi-Kanakkithodi, and Chiho Kim. Machine learning in materials informatics: Recent applications and prospects. *npj Computational Materials*, 3(1):54, 2017.